A visual guide to an ensemble classification study

100,000 records
eight input fields
one recorded outcome

What do models see?

Two different ways of reading the same eight signals. This study shows how Random Forest and Gradient Boosting turn a patient profile into an estimated likelihood — and where that estimate stops.

Try an example profile →

See the evidence first



The central idea

One profile. Two readings. Each score answers a question.

← Start here

Use an example profile or enter values from the eight fields used by the study.

See what both models make of one profile.

The same inputs go to both learners. You will see two estimated likelihoods, written in plain language. This is an educational demonstration, not a diagnosis.

Do not enter names or other identifying information. The example profile is illustrative and does not represent a real person.

The prefilled values are an illustrative example, not a real person. Each sample button fills a profile and runs both trained models.

About the person

Basic fields as they appear in the source dataset.

Gender

Female

Age years

45

Health history

Recorded history, not a complete medical history.

High blood pressure history

Heart disease history

Two estimates for the same profile.

The percentage is a model output, not a personal medical risk score.

Random Forest

6.6%

predicted probability in this demonstration

Lower likelihood

Both models read the same profile.

← Start here

A model can be "better" on one measure and less strong on another. That is why this page shows the questions behind the scores.

Which model reads better?

On this held-out test, the models are nearly tied overall. Gradient Boosting leads slightly on ROC-AUC, recall, and F1. Random Forest has a small precision and accuracy advantage.

Read the question first.

A metric is not a verdict. It is one way of asking how the model behaved on records it had not seen during training.

Measure	What it asks	Random Forest	Gradient Boosting
Accuracy	How often was the overall prediction correct?	0.972	0.972
Precision	When the model flagged a positive, how often was it right?	1.000	0.993
Recall	How many recorded positive cases did it find?	0.675	0.679
F1 score	A balance between precision and recall.	0.806	0.806
ROC-AUC	How well did it rank positive cases above negative ones?	0.976	0.979

These values are loaded from the deployed training record.

← Start here

The model can only learn from the fields that are supplied. The fields themselves are part of the story.

What can the model see?

Eight fields from a public diabetes prediction dataset. Understanding those fields is part of understanding the output.

Try the same profile →

1,00,000 records in the dataset	**20,000** held out for testing
8 input fields	**1** recorded binary outcome

Positive cases are less common in this dataset, so accuracy alone cannot tell the whole story. Precision, recall, F1, and ROC-AUC help us see the model from more than one angle.

Missing values

The pipeline includes median handling for numeric fields and the most common category for text fields. The final training file reports no missing values, but the safeguard stays in the pipeline.

Categories

Gender and smoking history become model-readable fields through one-hot encoding. Unknown categories are ignored safely at prediction time.

Numeric values

Extreme values